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POSTER

Requirements Traceability: A Systematic Literature Review

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ABSTRACT

Software developers employ requirements traceability to ensure that each requirement is tracked throughout the software development life cycle, providing a clear link between requirements and the final product. Despite its importance in software development, studies show that there are a number of challenges facing requirements traceability. In this paper, we provide a systematic review of requirements traceability methods and tool support covering 2013 – 2024 from the major databases. The findings show that methods and tools covering the software development life cycle are missing, and evaluations based on industrial case studies are meagre, which may lower the confidence in applying the proposed approaches in practice. Moreover, the paper highlights the need for developing requirements traceability methods supporting agile and DevOps practices.

CCS CONCEPTS

• **Software and its engineering → Requirements analysis**.

KEYWORDS

Requirements Traceability, Requirements Management, Requirements Engineering, Systematic Literature Review

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1 INTRODUCTION

Requirements traceability (RT) is a core activity of requirements management and has many functions, such as helping in identifying the source of a requirement, analyzing change impact, verification of test cases, and tracking project progress ([3, 8, 10]). Despite its benefits, studies highlight significant implementation challenges. While several reviews have synthesized research on requirements traceability, many focus on specific aspects such as link generation. However, the rapidly evolving landscape of software engineering practices and increasing complexity of modern software systems necessitate an up-to-date, and comprehensive review. Although previous reviews have provided insights into traceability methods and tools, they often lack thorough examination of the effectiveness of the proposed approaches, particularly regarding empirical validation and emerging technologies like blockchain. This study aims to address these gaps by providing a comprehensive review of existing methods and tools, and uncover areas requiring further research.

1.1 Review Questions

To achieve the objectives of the study, the following review questions have been formulated.

RQ1: What requirements traceability methods are available in the literature?
RQ2: What tool support has been provided for requirements traceability, and what key features do they offer?
RQ3: How is the effectiveness of the requirements traceability approaches evaluated?
RQ4: What open challenges (or limitations) have been identified concerning requirements traceability?

2 RESEARCH METHOD

Since we aim to provide a comprehensive review of existing literature on requirements traceability, systematic literature review (SLR) is selected as an appropriate method for our research. The SLR was carried out following the guidelines outlined in [4], which provides three steps: Planning the review, conducting the review, and reporting the review.

3 RESULTS

The search process stages were captured using Preferred Reporting Items for Systematic Reviews and Meta-Analyses(PRISMA) flow chart [6]. See Figure 1.

3.1 Methods - RQ1

Our analysis of the 55 studies show that RT methods focus on different RT activities. These methods varied in their scope and functionality, with some specifically targeting particular traceability relationships, such as requirements to test cases, requirements to design elements, and requirements to source code. The methods offered diverse features including RT modeling capabilities, automated trace link generation, requirements coverage assessment, project progress monitoring, change impact analysis tools, bi-directional traceability support, and general RT implementation guidelines. Our analysis also grouped studies by methodological similarities, examining empirical evaluation presence in each category. Information Retrieval (IR) methods dominated the research, primarily focusing on accuracy and effort improvement, with most including empirical evaluations. However, studies using emerging technologies like blockchain lack empirical validation, hindering

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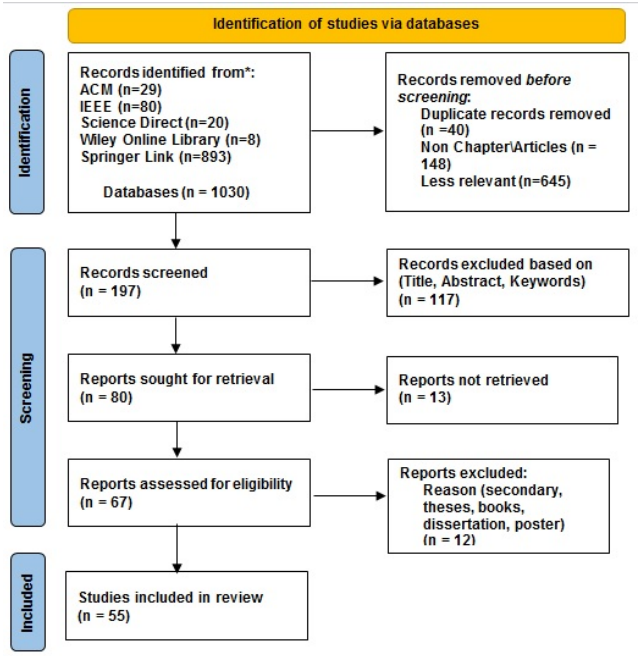

Figure 1: Flow Diagram for the Screening Process

their practical adoption. Wang [11] identified a critical issue: the absence of standardized benchmarks, with new methods typically compared against traditional baselines rather than state-of-the-art approaches, complicating accurate assessment of methodological improvements.

3.2 Tools - RQ2

Our review identified nine studies focusing on requirements traceability tools. The tools address specific aspects such as interdisciplinary project coordination, requirements-to-design traceability, and model-based solutions for UML integration. This narrow focus creates practical implementation challenges, requiring multiple tool integration for comprehensive coverage. The findings highlight a critical gap: the need for comprehensive tools supporting the full range of requirements traceability activities.

3.3 Effectiveness Measures - RQ3

Our analysis of evaluation metrics across 30 empirical studies revealed that 83% employed recall and precision measurements. This reflects the widespread use of information retrieval (IR) methods in requirements traceability, where these metrics are standard evaluation tools [7]. In the RT context, recall measures the percentage of correctly recovered links, while precision indicates the percentage of retrieved links that are correct or relevant.

3.4 Challenges - RQ4

Numerous studies have proposed solutions aimed at improving accuracy of existing techniques and recovery of traceability links, yet challenges of full automation and accuracy remain significant concerns. While Demi [1] proposed a blockchain solution for enabling communication, coordination, and trustworthiness between distributed stakeholders, their approach faced scalability issues, a challenge also highlighted by Marques [5], Hallerstede [2], and Sinha [9], who emphasized the need for empirical analysis with diverse datasets. Our analysis revealed four major persistent challenges: the lack of standardized evaluation benchmarks, insufficient empirical evaluation of emerging technology-based solutions (particularly blockchain), fragmented tools focusing on specific traceability activities rather than comprehensive solutions, and limited research addressing requirements traceability implementation in modern development methodologies such as agile and DevOps.

4 CONCLUSION

Our systematic review analyzed 55 studies examining methods and tools addressing requirements traceability challenges, and to what extent empirical evaluations are carried out for the proposed solutions. While requirements traceability offers significant benefits in product quality and project management, our analysis revealed that existing solutions face crucial limitations including lack of agility and scalability, limited applicability across different project types (particularly in agile and DevOps environments), insufficient empirical evaluation, and validation against traditional rather than state-of-the-art solutions. These findings highlight the need for next-generation requirements traceability solutions that are adaptable, scalable, accurate, automated, and applicable across diverse project types, with rigorous validation through both academic and industry evaluations.

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